

Indian Knowledge Systems (IKS)

Reference Guide to Scholars, Texts & Contributions

Module 1 — Foundations, Context & Critics

Colonial Education Policy

Thomas Babington Macaulay

Period/Era: 1800–1859 (Minute dated 1835)

Key Work(s): Minute on Indian Education (1835)

Contribution: Advocated replacing indigenous education with English-medium Western education to create a class of Indians 'Indian in blood and colour, but English in taste, in opinions, in morals, and in intellect.' Directly led to the marginalisation of Sanskrit/Persian learning and traditional pathashalas/gurukuls in favour of a British-style curriculum.

Exam Point: Always link Macaulay's Minute to (a) decline of indigenous institutions, (b) creation of a new clerical class, (c) rupture in continuity of IKS transmission.

Lord William Bentinck

Period/Era: Governor-General, 1828–1835

Key Work(s): English Education Act, 1835

Contribution: As Governor-General, approved and implemented Macaulay's recommendations, formally making English the medium of instruction and government business in India.

Exam Point: Pair with Macaulay — Macaulay wrote the Minute, Bentinck enacted it.

Dharampal

Period/Era: 1922–2006 (modern historian)

Key Work(s): The Beautiful Tree: Indigenous Indian Education in the Eighteenth Century (1983)

Contribution: Used British colonial survey records themselves to document that indigenous Indian education (pathashalas, gurukuls) was widespread, well-organised, and cut across caste lines before British disruption — a key modern counter-narrative to the 'India was uneducated' colonial claim.

Exam Point: Useful for the 'Need of revisiting Ancient Indian Traditions' sub-topic — shows why revisiting matters.

Ancient Educational Institutions

Panini (as an educational figure)

Period/Era: c. 6th–4th century BCE

Key Work(s): Ashtadhyayi

Contribution: Represents the rigour of oral pedagogical tradition (guru-shishya method) that trained students to memorise and internalise highly systematic grammar — often cited as an example of the sophistication of ancient pathashala/gurukul learning.

Local Heritage Site — Kanheri Caves

Kanheri Buddhist Monastic Community

Period/Era: 1st century BCE – 10th century CE

Key Work(s): Kanheri Cave complex, Mumbai (rock-cut cisterns)

Contribution: Anonymous Buddhist monk-engineers who designed one of the earliest known rainwater harvesting and water-management systems in India — over 100 rock-cut cisterns with channels to collect and store monsoon water for the monastery, an example of applied ancient hydrological engineering.

Exam Point: For local-site questions, mention the cistern network, gradient-based channel design, and its continued relevance to sustainable water management.

Module 2 — Core Disciplines

1. Medicine — Ayurveda

Charaka

Period/Era: c. 300 BCE – 200 CE (compiler)

Key Work(s): Charaka Samhita

Contribution: Foundational text of internal medicine (Kayachikitsa); systematised diagnosis, physiology, pharmacology, and medical ethics (the 'Charaka Oath', a precursor to the Hippocratic Oath).

Exam Point: Charaka = internal medicine/physician's text.

Sushruta

Period/Era: c. 6th century BCE (traditionally earlier)

Key Work(s): Sushruta Samhita

Contribution: Known as the 'Father of Surgery'; described over 300 surgical procedures, 120+ surgical instruments, plastic surgery (rhinoplasty), and cataract surgery.

Exam Point: Sushruta = surgery (Shalya Chikitsa).

Vagbhata

Period/Era: c. 7th century CE

Key Work(s): Ashtanga Hridayam / Ashtanga Sangraha

Contribution: Synthesised the works of Charaka and Sushruta into a unified, systematic treatise covering all eight branches of Ayurveda; became a widely used clinical manual.

Exam Point: Vagbhata = synthesiser of Charaka + Sushruta.

Dhanvantari

Period/Era: Mythological/divine figure

Key Work(s): Regarded as the god of Ayurveda

Contribution: Revered as the divine originator of Ayurvedic medicine in Indian tradition; temples and Dhanvantari Jayanti (Dhanteras) commemorate this figure.

Exam Point: Symbolic/mythological — cite as the 'deity of Ayurveda', not a historical author.

2. Alchemy — Rasashastra

Nagarjuna (Rasashastra tradition)

Period/Era: c. 10th–11th century CE

Key Work(s): Rasa Ratnakara, Rasendra Mangala

Contribution: Pioneered Rasashastra — the science of mercury and mineral-based medicine preparation; developed methods for detoxifying and processing metals (mercury, sulphur) for therapeutic use.

Exam Point: Distinguish this Nagarjuna from the Buddhist Madhyamaka philosopher of the same name.

Govinda Bhagavatpada

Period/Era: Traditionally associated with early medieval period

Key Work(s): Rasa Hridaya Tantra

Contribution: Important early contributor to systematising alchemical/Rasashastra processes and mercury purification methods.

3. Mathematics

Aryabhata

Period/Era: 476–550 CE

Key Work(s): Aryabhatiya (499 CE)

Contribution: Introduced the place-value system and a symbol for zero conceptually; computed the value of pi to four decimal places; explained the rotation of the Earth on its axis and eclipses scientifically.

Exam Point: Aryabhata = pi approximation + Earth's rotation.

Brahmagupta

Period/Era: 598–668 CE

Key Work(s): Brahmasphutasiddhanta

Contribution: First to treat zero as a number with defined mathematical rules (addition, subtraction rules involving zero); rules for solving quadratic equations.

Exam Point: Brahmagupta = rules of zero as a number.

Bhaskara I

Period/Era: c. 600–680 CE

Key Work(s): Aryabhatiya Bhashya, Mahabhaskariya

Contribution: Wrote the earliest known commentary on Aryabhata's work; devised an approximation formula for the sine function.

Bhaskara II (Bhaskaracharya)

Period/Era: 1114–1185 CE

Key Work(s): Lilavati, Bijaganita, Siddhanta Shiromani

Contribution: Comprehensive works on arithmetic (Lilavati), algebra (Bijaganita), and astronomy; anticipated concepts resembling infinitesimal calculus and differential calculus centuries before Newton/Leibniz.

Exam Point: Bhaskara II = Lilavati (arithmetic) + early calculus-like ideas.

Madhava of Sangamagrama

Period/Era: c. 1340–1425 CE

Key Work(s): Founder of the Kerala School of Astronomy and Mathematics

Contribution: Developed infinite series expansions for trigonometric functions (sine, cosine, arctangent) — precursor to the Taylor/Madhava series, predating European calculus by two centuries.

Exam Point: Madhava = infinite series / Kerala School founder.

Mahavira (mathematician)

Period/Era: 9th century CE

Key Work(s): Ganita Sara Sangraha

Contribution: Jain mathematician who wrote a comprehensive treatise on arithmetic operations, fractions, and permutations/combinations.

Varahamihira

Period/Era: 505–587 CE

Key Work(s): Pancha Siddhantika, Brihat Samhita

Contribution: Synthesised five earlier astronomical treatises; contributed to trigonometry and astronomy; Brihat Samhita also touches architecture and agriculture.

Exam Point: Cross-links to Krishi Shastra and Architecture topics too.

4. Logic — Nyaya

Gautama (Akshapada Gautama)

Period/Era: c. 6th century BCE

Key Work(s): Nyaya Sutras

Contribution: Founder of the Nyaya school of logic; established the framework of 16 categories (padarthas) including means of valid knowledge (pramana) — perception, inference, comparison, testimony.

Exam Point: Gautama = founder of Nyaya, 4 pramanas.

Vatsyayana (Nyaya commentator)

Period/Era: c. 4th–5th century CE

Key Work(s): Nyaya Bhashya

Contribution: Wrote the earliest surviving major commentary on the Nyaya Sutras, clarifying and expanding its logical framework.

Exam Point: Do not confuse with the author of the Kamasutra (same name, different scholar).

Gangesha Upadhyaya

Period/Era: c. 12th–13th century CE

Key Work(s): Tattvachintamani

Contribution: Founder of Navya-Nyaya ('New Logic'), which refined logical and epistemological analysis with precise technical language, influencing Indian philosophy for centuries.

Exam Point: Gangesha = Navya-Nyaya (New Logic).

Dignaga

Period/Era: c. 480–540 CE

Key Work(s): Pramanasamuccaya

Contribution: Buddhist logician who systematised Buddhist epistemology and theory of inference, greatly influencing later Indian logic (including Nyaya responses to it).

Dharmakirti

Period/Era: c. 6th–7th century CE

Key Work(s): Pramanavarttika

Contribution: Advanced Buddhist logic and epistemology after Dignaga; refined theories of perception and inference.

5. Art of Governance — Arthashastra

Chanakya (Kautilya / Vishnugupta)

Period/Era: c. 4th century BCE

Key Work(s): Arthashastra

Contribution: Comprehensive treatise on statecraft, economic policy, military strategy, taxation, and diplomacy; architect of the Mauryan empire under Chandragupta Maurya; introduced the Mandala theory of foreign policy

(concentric circles of ally/enemy states).

Exam Point: Chanakya = Arthashastra + Mandala theory + taxation/economic policy — appears across Governance, Strategic Studies, Taxation, and Trade topics.

Kamandaka

Period/Era: c. 4th–8th century CE (dating debated)

Key Work(s): Nitisara

Contribution: Condensed and versified the political principles of Kautilya's Arthashastra into an accessible text on statecraft and ethics of kingship.

Exam Point: Kamandaka = 'summary/poetic version' of Arthashastra ideas.

Module 3 — Elective Topics (Select Any Five)

Aesthetics

Bharata Muni

Period/Era: c. 200 BCE – 200 CE

Key Work(s): Natyashastra

Contribution: Founding text of Indian aesthetics and dramaturgy; introduced the Rasa theory (nine rasas/emotions) as the foundation of artistic experience across drama, dance, and music.

Exam Point: Bharata Muni links Aesthetics AND Natyashastra topics — core figure for both.

Abhinavagupta

Period/Era: c. 950–1016 CE

Key Work(s): Abhinavabharati (commentary on Natyashastra), Dhvanyaloka-Lochana

Contribution: Synthesised Rasa theory with Dhvani (suggestion) theory, giving Indian aesthetics its most sophisticated philosophical treatment; also a major Kashmir Shaivism philosopher.

Anandavardhana

Period/Era: 9th century CE

Key Work(s): Dhvanyaloka

Contribution: Propounded the Dhvani (theory of suggestion) school of literary criticism, arguing suggested meaning is superior to literal meaning in poetry.

Town Planning

Kautilya (Chanakya)

Period/Era: c. 4th century BCE

Key Work(s): Arthashastra (sections on Durga/fort and city planning)

Contribution: Prescribed detailed principles for fortified city layout, roads, drainage, and zoning of residential/commercial/administrative areas.

Sage Maya (attributed)

Period/Era: Traditional/legendary attribution

Key Work(s): Mayamata

Contribution: Attributed authorship of one of the key Vastu Shastra texts governing town and temple layout, proportion, and orientation.

Exam Point: Mayamata + Manasara = twin classical town-planning/architecture treatises.

Strategic Studies

Chanakya (Kautilya)

Period/Era: c. 4th century BCE

Key Work(s): Arthashastra

Contribution: Developed the Mandala (circle of states) theory and Saptanga (seven limbs of state) theory, foundational to Indian strategic and international-relations thought.

Exam Point: Repeat figure — emphasise Mandala theory specifically here vs. taxation elsewhere.

Kamandaka

Period/Era: c. 4th–8th century CE

Key Work(s): Nitisara

Contribution: Elaborated strategic and diplomatic principles (war, peace, alliance) building on Kautilyan thought.

Krishi Shastra (Agriculture)

Parashara (agricultural author)

Period/Era: Dating uncertain, ancient

Key Work(s): Krishi Parashara

Contribution: One of the earliest dedicated treatises on agriculture — covers soil types, seasons, rainfall prediction, seed selection, and cattle care.

Exam Point: Distinguish from the sage Parashara of the Mahabharata/astrology tradition.

Kashyapa (agricultural author)

Period/Era: Dating uncertain, ancient/medieval

Key Work(s): Kashyapiyakrishisukti

Contribution: Comprehensive agricultural treatise covering irrigation, manure, plant diseases and remedies, and horticulture.

Varahamihira

Period/Era: 505–587 CE

Key Work(s): Brihat Samhita (agriculture-related chapters)

Contribution: Included sections on rainfall prediction, groundwater detection, and plant treatment within his broader astronomical/astrological encyclopaedia.

Vyakarana & Lexicography

Panini

Period/Era: c. 6th–4th century BCE

Key Work(s): Ashtadhyayi

Contribution: Composed the most systematic grammar of any classical language using ~4,000 sutras (rules) with a formal, algebra-like notation; standardised classical Sanskrit and influenced modern computational linguistics.

Exam Point: Panini = single most important name for Vyakarana AND Linguistics topics.

Katyayana (grammarian)

Period/Era: c. 3rd century BCE

Key Work(s): Varttikas on Ashtadhyayi

Contribution: Wrote critical annotations (varttikas) on Panini's sutras, refining and extending Paninian grammar.

Patanjali (grammarian)

Period/Era: c. 2nd century BCE

Key Work(s): Mahabhashya

Contribution: Wrote the definitive commentary on Panini and Katyayana, called the 'Great Commentary'; also discusses philosophy of language.

Exam Point: This grammarian-Patanjali is traditionally treated as distinct from the Yoga Sutras' Patanjali, though some traditions identify them as the same person — mention both views if asked.

Amarasimha

Period/Era: c. 4th–5th century CE (Gupta period)

Key Work(s): Amarakosha

Contribution: Composed the most influential Sanskrit thesaurus/lexicon, organising vocabulary by subject in verse form for ease of memorisation.

Exam Point: Amarasimha = lexicography (dictionary), not grammar rules.

Natyashastra

Bharata Muni

Period/Era: c. 200 BCE – 200 CE

Key Work(s): Natyashastra

Contribution: Comprehensive treatise covering drama, dance, music, stagecraft, and the Rasa theory of aesthetic emotion — the foundational text of Indian performing arts.

Abhinavagupta

Period/Era: c. 950–1016 CE

Key Work(s): Abhinavabharati

Contribution: Most authoritative commentary on the Natyashastra, deepening the philosophical basis of Rasa theory.

Ancient Sports

King Someshvara III (Someshvara Chalukya)

Period/Era: Ruled c. 1126–1138 CE

Key Work(s): Manasollasa

Contribution: Royal encyclopaedic text covering statecraft, arts, and a notable section on games, sports, and physical culture (wrestling, swimming, martial exercises) practised in the 12th century.

Exam Point: Manasollasa = key text if 'Ancient Sports' is chosen — it's an encyclopaedic royal text, not sport-specific.

Yoga and Wellbeing

Patanjali (Yoga)

Period/Era: Dating debated, c. 2nd century BCE – 4th century CE

Key Work(s): Yoga Sutras

Contribution: Systematised yoga philosophy into the Ashtanga Yoga (eight-limbed path): Yama, Niyama, Asana, Pranayama, Pratyahara, Dharana, Dhyana, Samadhi.

Exam Point: Patanjali (Yoga) = Ashtanga Yoga's eight limbs — a favourite exam question.

Swami Vivekananda

Period/Era: 1863–1902

Key Work(s): Raja Yoga (1896)

Contribution: Introduced and popularised yoga philosophy (particularly Raja Yoga) to the West after his 1893 Chicago address, sparking the modern global yoga movement.

Tirumalai Krishnamacharya

Period/Era: 1888–1989

Key Work(s): Yoga Makaranda

Contribution: Known as the 'father of modern yoga'; revived and systematised Hatha Yoga asana practice, training influential students (K. Pattabhi Jois, B.K.S. Iyengar) who globalised distinct yoga styles.

Linguistics

Panini

Period/Era: c. 6th–4th century BCE

Key Work(s): Ashtadhyayi

Contribution: Pioneered formal, rule-based descriptive linguistics millennia before modern linguistics; his generative-rule approach is cited as a precursor to computational grammar models.

Bhartrihari

Period/Era: c. 5th century CE

Key Work(s): Vakyapadiya

Contribution: Developed the philosophy of language, notably the Sphota theory — the idea that meaning is grasped as an indivisible flash of understanding rather than built word-by-word.

Exam Point: Bhartrihari = Sphota theory (philosophy of language, distinct from Panini's grammar rules).

Chitrasutra

Chitrasutra tradition (Vishnudharmottara Purana)

Period/Era: Compiled c. 5th–7th century CE

Key Work(s): Chitrasutra section of the Vishnudharmottara Purana

Contribution: Earliest surviving Indian treatise systematically dealing with the theory and practice of painting — proportion, perspective, categories of representation, and the aesthetic purpose of art. No single named individual author survives; it is a codified textual tradition.

Architecture

Vishwakarma (attributed/mythological)

Period/Era: Traditional/divine attribution

Key Work(s): Attributed source of Vastu Shastra knowledge

Contribution: Revered in tradition as the divine architect-craftsman; treatises on architecture are symbolically traced back to his knowledge.

Sage Maya / Manasara (attributed authors)

Period/Era: Traditional attribution, texts compiled over centuries

Key Work(s): Mayamata, Manasara

Contribution: Classical Vastu Shastra texts prescribing rules for temple and building design, proportion, site-selection, and orientation — foundational for Indian temple architecture (e.g., temple management traditions like Ambarnath).

Exam Point: Directly relevant to your syllabus's local-site example: Ambarnath Temple management.

Taxation

Kautilya (Chanakya)

Period/Era: c. 4th century BCE

Key Work(s): Arthashastra (taxation chapters)

Contribution: Laid out a detailed and graded taxation system (land revenue, trade tariffs, tolls) with principles of fairness and state welfare — an early treatise on public finance.

Banking

Hundi system practitioners (Shroffs/Shreni bankers)

Period/Era: Ancient to medieval period (institutional, no single founder)

Key Work(s): Referenced in Arthashastra and later medieval trade records

Contribution: Indigenous credit instrument (Hundi) and banking-like functions were carried out by merchant/banker guilds (Shreshthins, Shroffs); Kautilya's Arthashastra also discusses state treasury management and interest rates.

Exam Point: No single named 'banker' — answer via institutional Hundi/Shreni system + Kautilya's treasury principles.

Trade and Commerce

Kautilya (Chanakya)

Period/Era: c. 4th century BCE

Key Work(s): Arthashastra (trade regulation chapters)

Contribution: Regulated trade practices, weights and measures, customs duties, and merchant conduct as part of state economic policy.

Shreni (Guild) system (institutional)

Period/Era: c. 6th century BCE onward

Key Work(s): Referenced across Buddhist Jataka tales, inscriptions, and Arthashastra

Contribution: Merchant and craft guilds (Shrenis) functioned as self-regulating trade bodies, setting quality standards, resolving disputes, and even issuing credit — precursors to modern trade associations.

Note: For Module 3, only five topics are required per the syllabus — use this guide to select and go deep on the five you are examined on.